Markups and Public Procurement: Evidence from Czech Construction Tenders

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Excerpt: introduction, data, estimation under unconfoundedness, and conclusion

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Abstract

This paper analyzes the effect of public procurement on firm markups, using a panel dataset of Czech construction firms from 2006 to 2021. Markups are estimated through a structural framework, while biases are addressed using a selection on observables design and models for unobserved factors. Propensity score-based estimations indicate that firm markups increase by approximately 15% during contract years. A temporal analysis, employing synthetic control and matrix completion methods, reveals that treatment effects decline from around 30% in 2006 to 10% in 2021. These patterns are consistent with institutional improvements in the Czech Republic and offer empirical evidence of increasing efficiency in public spending.

Keywords: Models with Panel Data; Firm Behavior: Empirical Analysis; Procurement; Construction

JEL: C23, D22, H57, L74

[1]Predoctoral Fellow, Tobin Center for Economic Policy, Yale University (marek.chadim@yale.edu; https://marek-chadim.github.io). This paper is my MSc thesis, submitted to the Department of Economics, Stockholm School of Economics (defended December 2024, awarded the highest grade) and awarded First Place in the 32nd Young Economist of the Year competition of the Czech Economic Society, 2025. I am grateful to my thesis supervisor, Jaakko Meriläinen, for invaluable guidance; to Matěj Bajgar, Mitch Downey, and David Schönholzer for their helpful feedback; and to Jiří Skuhrovec of Datlab s.r.o. for providing administrative data on government procurement. This thesis builds on my earlier work at the Institute of Economic Studies, Charles University Prague (https://dspace.cuni.cz/handle/20.500.11956/184831), where I accessed firm financial statement data via MagnusWeb. Replication files: https://github.com/marek-chadim/thesis. This 10-page version reproduces, verbatim, the introduction, the data, the estimation under unconfoundedness (Section 3.2), and the conclusion (pages 2–3, 7, 15–19, and 32 of the thesis). The complete thesis is in the Stockholm School of Economics library: https://hhs.primo.exlibrisgroup.com/permalink/46SSOE_INST/hejhp1/alma991001619398206056.

1 Introduction

Public procurement constitutes approximately 12% of GDP in OECD countries, representing a substantial share of government expenditure (OECD, 2021). Ensuring its efficiency is vital for reducing resource waste and maximizing taxpayer value. Existing literature has examined key issues in public procurement, including discretion, political favoritism, and limited competition. However, much of this research has focused on comparing tenders, with relatively little attention devoted to differences between public procurement and private market outcomes.

This paper fills this gap by constructing a novel dataset and proposing methodological solutions to the identification challenges inherent in studying the dynamics of firm markups as firms enter and exit public procurement markets. To the best of my knowledge, this is the first study to provide credible econometric evidence on the relationship between markups and participation in public procurement. The findings contribute to ongoing debates on public procurement efficiency, market competition, and the implications of government contracts for firm-level market power.

Using the methodology of De Loecker and Warzynski (2012), I estimate markups under the assumption that cost-minimizing producers optimize variable input expenditures. This approach provides plant-level markup estimates without relying on assumptions about firms' competitive behavior in product markets. The results document the evolution of aggregate markups, showing a decline from 40% above marginal cost in 2006 to 30% in 2021, driven primarily by firms at the upper end of the markup distribution.

Next, I estimate the public procurement premium using non-experimental methods under the assumption of unconfoundedness, conditioning on information available prior to firms' engagement in public procurement. To mitigate biases arising from observational characteristics, I apply methodological advancements from Imbens and Xu (2024). Placebo tests, which compare the average markups of treated and control firms before the treated firms receive contracts, and sensitivity analyses benchmarking the omitted variable bias further reinforce the robustness of the findings.

Finally, I employ panel data models for binary interventions in longitudinal settings to estimate the causal effect of public procurement on the markups of government contractors, drawing on Arkhangelsky and Imbens (2024). These methods extend traditional difference-in-differences and two-way fixed-effects models by incorporating factor models, interactive fixed-effects models, and synthetic control approaches. The analysis emphasizes heterogeneity in causal effects and uses valid inference techniques suited to cases with a small number of treated units.

Drawing causal inferences from observational data, even with advanced methods, requires strong assumptions. The lack of natural experiments, instrumental variables, or regression discontinuity designs limits the availability of exogenous variation in this study. To address these challenges, I use a selection-on-observables approach, framed within the potential outcomes framework, to model counterfactual markups for government contractors in the absence of public procurement. The stable unit treatment value assumption (SUTVA) excludes interference between units and the dynamic effects of past treatments and outcomes. However, in competitive public procurement markets, violations of SUTVA can complicate the interpretation of causal effects. Despite this, the study prioritizes internal validity through a design-based balanced panel approach while enhancing generalizability by leveraging the full dataset. Balanced panel estimates, based on selection on observables, indicate that procurement raises contracting firms' markups by an average of 15%, with the full panel data model showing a decline in this effect from 30% in 2006 to 10% in 2021.

Industrial organization models with heterogeneous producers and firm specific markups provide potential explanations for these findings. More productive firms may absorb the costs associated with entering public procurement markets, which could explain why government contractors typically exhibit higher markups. Additionally, contractors may charge higher markups if they supply higher-quality goods produced using higher-quality inputs. Another plausible explanation is that procurement contracts often provide firms with long-term clients, leading to less elastic realized demand and enabling higher markups.

The literature on public procurement also highlights other mechanisms, such as discretion and political favoritism. My findings suggest that the public procurement markup premium has declined over time, consistent with reforms targeting single-bidding practices Titl (2023) and increased monitoring associated with European Union co-financing (Baránek and Titl, 2024).

The remainder of the paper is structured as follows. Section 1.1 reviews related work. Section 2 introduces the dataset and the construction of the public procurement indicator. Section 3.1 documents the distribution of markups and their descriptive relationship with public procurement. Section 3.2 presents the causal results under the unconfoundedness assumption, while Section 3.3 reports findings from panel data analyses. Methodological details supporting these sections are outlined in Appendices A, B, and C. Finally, Section 4 concludes the paper.

2 Data

I utilize a dataset obtained through licensed[2] access to the MagnusWeb database[3], which provides full company accounts for an unbalanced panel of 1,297 firms active in the Czech construction sector. The dataset contains 7,261 observations covering the period from 2006 to 2021. The sample size accounts for the markup estimation requirement of having at least one lag of data. For robustness of the production function estimates, the top and bottom percentiles of firms are trimmed based on the sales-to-cost-of-goods-sold ratio.

In the Czech Republic, public procurement contracts are awarded under nationwide regulations requiring the publication of contract details by procurers. The dataset includes contracts awarded by central, regional, and municipal governments, as well as government-owned enterprises. A private company[4] corrected and cleaned the raw data to ensure its accuracy. I link firm financial statements with public procurement records to construct a binary indicator of government tender sales. This classification identifies firms that operate exclusively in the private sector, those that enter public procurement, those that exit it, and those that remain as government contractors.

Year	No. Firms	Share Contractors
2006	227	0.38
2007	290	0.33
2008	348	0.32
2009	412	0.32
2010	497	0.28
2011	506	0.26
2012	457	0.33
2013	235	0.41
2014	243	0.40
2015	245	0.48
2016	338	0.51
2017	660	0.57
2018	708	0.55
2019	764	0.53
2020	769	0.55
2021	562	0.53
Total	7,261	0.44

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[3]Dun & Bradstreet Czech Republic, a.s.
[4]Datlab s.r.o.

3.2 Estimation of the Public Procurement Effect Under Unconfoundedness

In this section, I reanalyze data from the Czech construction sector, building on the structural inference of the underlying markup distribution, to evaluate the effect of public procurement engagement between 2010 and 2021 on firm pricing power. The primary outcome of interest is the annual contract year markup. I use multiple methods to estimate the causal effects of public procurement engagement under unconfoundedness. I adopt the potential outcomes framework, relying on two key assumptions—unconfoundedness and overlap—and address their plausibility using placebo and sensitivity analyses. [10]

The analysis focuses on a binary treatment, examining the efficiency of government tenders in the Czech construction sector compared to private markets during 2006–2021. A key challenge is the lack of detailed information on the treatment assignment mechanism. However, the availability of lagged outcomes allows for validation of the unconfoundedness assumption. The analysis is restricted to a balanced panel as a design-based decision. This approach ensures consistency over time and facilitates intuitive matching procedures, though it limits the analysis to 26 firms observed over the full 16-year period.

The control group includes 161 firm-year observations from firms without any government tender revenues between 2010 and 2021. The treatment group includes 151 firm-year observations from firms receiving government construction tender payments between 2010 and 2019. Treated firms include both recurring contractors and one-time providers. The total contract value amounts to 25.9 billion Czech Crowns in 2021, equivalent to approximately 1.2 billion USD based on the 2021 exchange rate (1 USD = 21.5 CZK).

	2010	**2011**	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**Total**
Contract Value (Billions CZK)	2.20	0.27	0.60	2.82	1.11	0.84	0.59	1.99	5.05	3.84	2.98	3.60	25.90
Number of Contracting Firms	7	8	9	13	11	14	12	17	17	15	15	13	151

Engagement in government contracts is not randomly assigned and cannot ensure comparability between treatment and control groups at the time of public procurement entry. However, I present results under the assumption that unconfoundedness holds when conditioned on a set of observable covariates, including the year in which the contract was awarded. Notably, I have data on past markups spanning 4 years (2006–2009) to 15 years (2006–2020) before a firm won a contract. These historical outcomes are used as either conditioning variables or placebo outcomes.

[10] See Methods B.

In the subsequent analysis, I use the natural logarithm of markups from three post-contract periods as the outcome variables, denoted as $Y_{i,0}, Y_{i,1}, Y_{i,2}$, where $t = 0$ represents the year the contract was awarded. The log markups from the three years immediately preceding the contract win $Y_{i,-3}, Y_{i,-2}, Y_{i,-1}$ and their average will serve as placebo outcomes. Additionally, the log markups from the three years preceding the placebo period $Y_{i,-6}, Y_{i,-5}, Y_{i,-4}$ will be included as covariates for adjustment. This adjustment will also incorporate time-invariant and pre-contract variables such as financial statement data—specifically, sales, cost of goods sold, assets, and number of employees—as well as the contract year and the NACE 2-digit division. Lagged values of the public procurement indicator are included to compare firms with similar pre-treatment histories.

Table 1: Pretreatment Summary Statistics

Observations Stratified by Engagement in Public Procurement	$\{i, t; W_{it} = 0\}$ N=161 **mean (s. d.)**	$\{i, t; W_{it} = 1\}$ N=151 **mean (s. d.)**	**diff / sd**
No. Employees	52.57 (42.03)	104.96 (61.67)	0.993
$\ln(\text{Sales})_{t-4}$	18.52 (1.01)	19.27 (0.85)	0.799
$\ln(\text{Costs})_{t-4}$	17.89 (0.95)	18.52 (0.82)	0.713
$\ln(\text{Markup})_{t-4}$	0.24 (0.13)	0.34 (0.17)	0.640
$\ln(\text{Assets})_{t-4}$	16.05 (1.68)	16.97 (1.37)	0.603
NACE 42	0.04 (0.20)	0.11 (0.32)	0.259
NACE 43	0.40 (0.49)	0.44 (0.50)	0.081

Note: I report observations from firms in a strongly balanced panel, excluding 26 firms already active in public procurement in 2006. The period from 2006 to 2009 is unavailable due to matching and placebo requirements of at least four years. This setup results in $N \times T = 312$ observations during 2010–2021. The measure diff/sd represents the absolute mean difference, standardized by the pooled standard deviation:

$$\text{diff/sd} = \left|\bar{X}_0 - \bar{X}_1\right| \left(\sqrt{(s_0^2 + s_1^2)/2}\right)^{-1}.$$

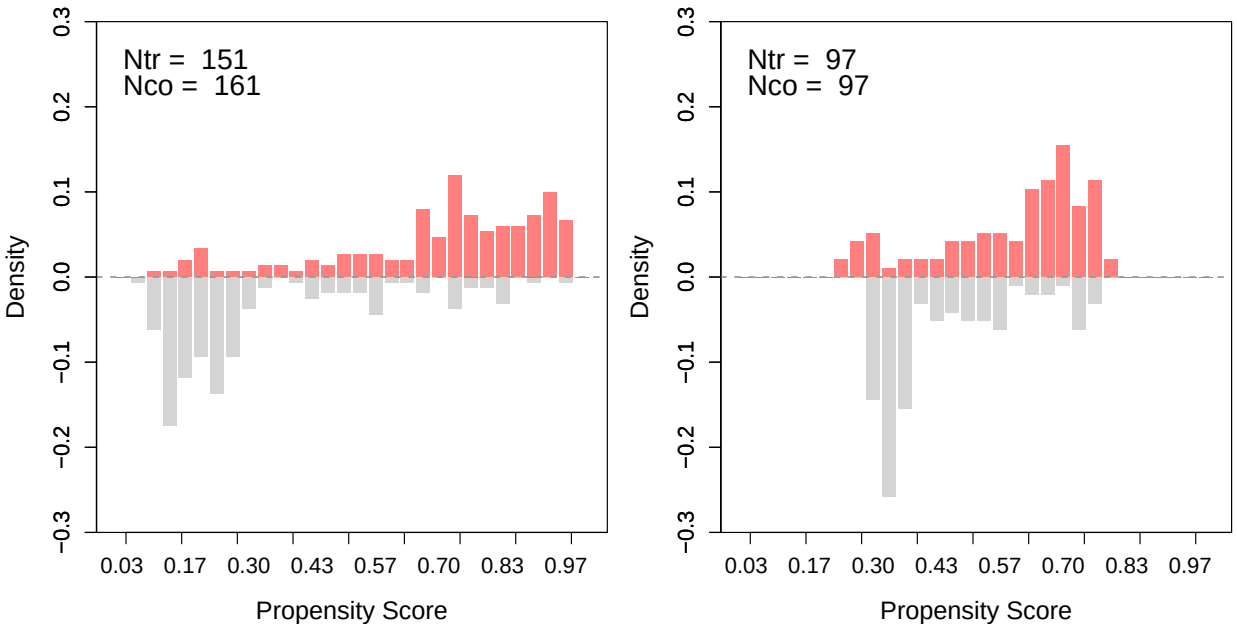
In Table 1, I present the averages and standard deviations for the observations used in the estimation sample. The sample includes firms observed over the full 16-year period, excluding those engaged in public procurement as of 2006. Additionally, I condition the sample on four lagged observations of the pre-treatment variable.

The final dataset consists of 26 firms tracked across 10 periods of instantaneous treatment effects between 2010 and 2021. This includes 161 firm-year observations for firms exclusively active in the private sector ($i, t; W_{it} = 0$) and 151 for firms engaged in government contracts ($i, t; W_{it} = 1$). Average covariate values by treatment status are reported, normalized by their respective standard deviations.

Table 1 shows that the baseline difference in average markups between treated firms (future contractors) and control firms (those active only in the private sector during 2010–2021) amounts to 0.66 standard deviations. I do not report the t-statistic for this difference, as it partially reflects sample size. Larger normalized differences clearly indicate overlap issues, with values exceeding 0.25 standard deviations generally considered substantial (Imbens and Wooldridge, 2009).

To estimate the treatment effects, the controls are used to calculate the conditional mean, $\mu_0(x) \equiv [Y_i(0)|X_i = x]$, which is then used to predict the missing control outcomes for the treated units. The significant disparity in covariate distributions between the two groups—up to 0.99 standard deviations from zero—suggests that linear regression may rely heavily on extrapolation, making results highly sensitive to the model's functional form.

Figure 3: Assessing Overlap

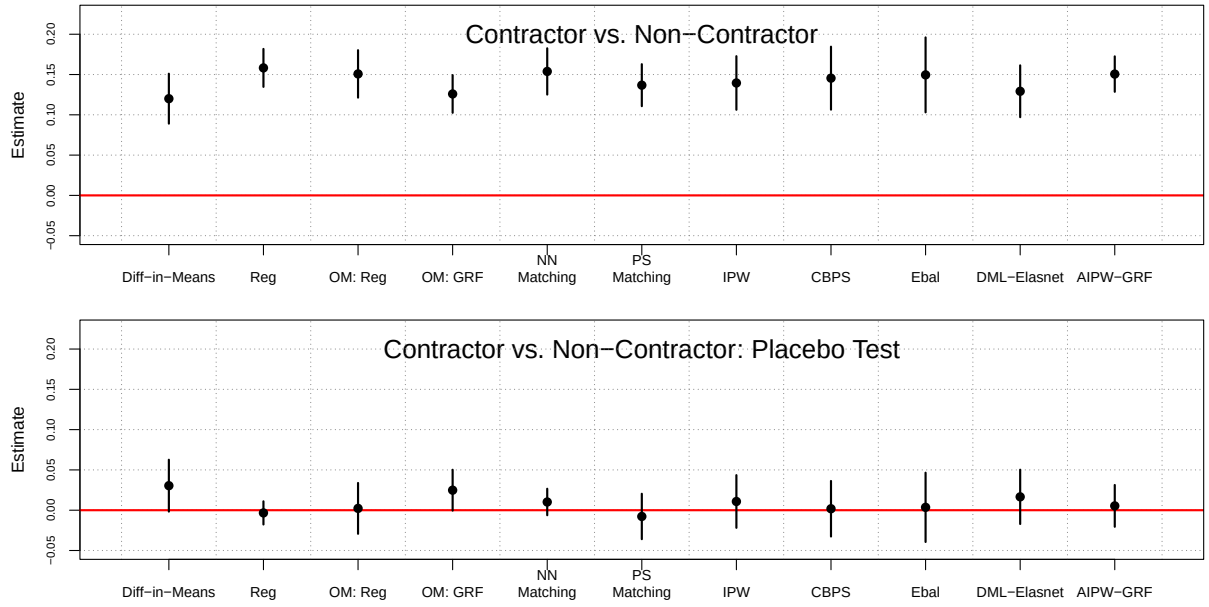


(a) Full sample (b) Trimmed sample

Note: Propensity scores estimated through Generalized Random Forest and re-estimated after trimming in (b).

Figure 3 evaluates the overlap between the treatment and control groups using the covariates described earlier. The figure shows that while the propensity score distribution for individuals in the treatment group differs from that of the control group, the propensity scores of the treatment group remain within the range of the control group's distribution. To improve overlap, I exclude observations with propensity scores above 0.8, removing 8 control units and 53 contractors. I refine the control group using 1:1 propensity score matching without replacement.

Figure 4: ATT Given Unconfoundedness and Placebo Estimates



Note: The top figure displays ATT (percentage) estimates comparing government contractors to non-contractors for the average annual log markups in Year 0, with 95% confidence intervals. The bottom figure shows ATT estimates for the placebo outcome (average markups from Year –3 to Year –1), with 95% confidence intervals. This analysis employs eleven estimators, including difference-in-means, linear regression, Oaxaca Blinder, GRF as an outcome model, 1:5 nearest neighbor matching with bias correction, propensity score matching, IPW with propensity scores estimated by GRF, CBPS, entropy balancing, double/debiased machine learning with elastic net (using `DoubleML`), and AIPW with GRF (using `grf`).

Table 2: ATT Given Unconfoundedness and Placebo Estimates

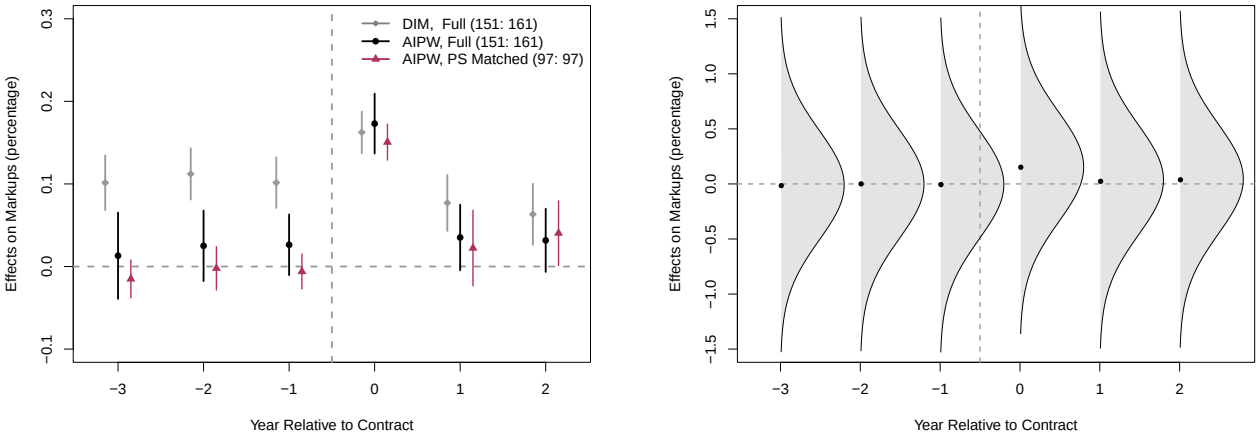
Effect on Markups	**Contract**	**Pre-Contract Average**
Difference-in-Means	0.12 (0.02)	0.03 (0.02)
Regression	0.16 (0.01)	-0.00 (0.01)
Oaxaca Blinder	0.15 (0.01)	0.00 (0.02)
GRF	0.13 (0.01)	0.03 (0.01)
NN Matching	0.15 (0.01)	0.01 (0.01)
PS Matching	0.13 (0.01)	-0.00 (0.01)
IPW	0.14 (0.02)	0.01 (0.02)
CBPS	0.15 (0.02)	0.00 (0.02)
Entropy Balancing	0.15 (0.03)	-0.00 (0.02)
DML-ElasticNet	0.16 (0.01)	-0.01 (0.01)
AIPW-GRF	0.15 (0.01)	0.00 (0.01)

Note: ATT estimates for the real outcome (markups in Year 0) and the placebo outcome (average markups from Year –3 to Year –1) are based on the trimmed dataset, excluding extreme propensity scores (above 0.8). Robust standard errors are in parentheses. The outcome is measured in natural logarithms and presented as percentage effects. These estimates are visualized in Figure 4.

Figure 4 displays the ATT estimates from various estimators for the real outcome (log markup in Year 0) and the placebo outcome (average log markups from Year -3 to Year -1). The figure demonstrates that different covariate adjustment methods yield consistent results, extending the descriptive results from the previous section: participation in public procurement leads to a significant increase in markups, with an average increase of up to 15% across methods. Applying doubly robust estimators, such as AIPW-GRF, yields nearly zero estimates for the placebo outcomes, further supporting the unconfoundedness assumption.

I estimate the ATT and CATT for log markups from Year -3 to Year 3, using both difference-in-means and AIPW-GRF methods. These results are presented in Figure 5. This figure resembles an event study plot often used in panel data analysis; however, the primary identification assumption here is unconfoundedness. The estimates from AIPW-GRF and difference-in-means differ substantially, with AIPW-GRF producing more credible results. The difference-in-means approach performs poorly in the placebo tests, while AIPW-GRF yields placebo estimates that are nearly zero. These findings provide strong corroborative evidence from the placebo tests: the CATT estimates are close to zero in the pre-contract years and indicate minimal treatment effect heterogeneity among government contractors.

Figure 5: ATT and CATT Estimates



(a) ATT (b) CATT

Note: Figures show the ATT and CATT estimates. The outcome variables include markups from 3 years before winning to 6 years after winning (Years –3 to 6). Estimates for pre-contract outcomes (Years –3 to –1) are used as placebo tests, validating the unconfoundedness assumption.
Subfigure A displays ATT estimates using the difference-in-means estimator (gray diamonds) and the AIPW-GRF estimator (black solid circles for the original data and red triangles for the trimmed data).
Subfigure B shows the distribution of CATT estimates using AIPW-GRF for the trimmed sample, with black dots representing the corresponding ATT estimates.

4 Conclusion

Using firm-level data on publicly traded firms in the Czech construction sector, this study employs a method that accommodates diverse price-setting strategies and nonconstant returns to scale to estimate firm-specific markups. It describes the evolution of market power through the distributional characteristics of markups. Between 2006 and 2021, markups declined from 40% to nearly 30%, a decline primarily driven by firms with the highest initial markups. Over time, the distribution of markups became less skewed, with a thinner upper tail.

The decline in Czech construction aggregate markups is linked to the dynamics of the public procurement markup premium. The relationship is validated using policy evaluation designs to account for both observable and unobservable selection bias.

In the balanced sample, the year-of-contract-treatment effect is estimated to be 15% under selection on observable lagged variables, while the same-sample fixed-effects analysis yields a point estimate of 20%. Using both the unconfoundedness approach and the TWFE model to test robustness to alternative identifying assumptions leads to the bracketing property bounds on the estimated causal effect (Angrist and Pischke, 2009, pp. 185).

The analysis also integrates Synthetic Difference-in-Differences re-weighting and matching on pre-exposure trends to reduce reliance on parallel-trend assumptions. This approach accounts for additive unit-level shifts while utilizing data-driven synthetic control methods.

Finally, the full sample estimation employs interactive fixed effects and matrix completion methods. The framework is validated using diagnostic tests of out-of-sample predictions for untreated potential outcomes, which are robust to model misspecification and overfitting. Subgroup analysis indicates that the overall effect is primarily driven by government contracts awarded in the early years of the sample. The estimated average public procurement effect on government contractors is 30% in 2006 and decreases to 10% by 2021.

These findings contribute to the public procurement literature by establishing a relationship between markups and public procurement. They also provide a benchmark for assessing the competitiveness of government projects relative to those in the private sector. The results provide empirical support for reforms aimed at eliminating single-bidding practices and increasing transparency in the tendering process that prevents favoritism in public procurement. They offer policy-relevant insights, highlighting the role of institutional improvements in enhancing the efficiency of public spending.